Juniper JUNOS Core Node Failure SOP: Hardware/Environmental Fault Resolution for Node BLR-CORE-01

Node: BLR-CORE-01 | **Vendor:** Juniper | **Severity:** Warning | **Alarm Code:** ALARM_4264

Description

The failure of Node BLR-CORE-01 due to hardware/environmental faults (ASIC memory parity errors, line card crashes, power supply redundancy loss) is a critical event that can impact the stability and performance of the Juniper JUNOS core network. This SOP is designed to provide a systematic approach to identifying and resolving the root cause of the failure, leveraging the Juniper JUNOS CLI for diagnostic and resolution purposes. The failure domain is primarily attributed to the Line Card (LC) and the Presence Detection Module (PDM) in the Juniper JUNOS chassis, which are prone to ASIC memory parity errors and power supply redundancy loss due to environmental factors such as temperature fluctuations, humidity, and power supply instability. The Juniper JUNOS core node's robust architecture and redundant design ensure that the network remains operational even in the event of a hardware failure. However, prompt resolution of the issue is crucial to prevent network degradation and potential outages.

Pre-requisite Checks

- Step: Verify the site location and access permissions
- Step: Confirm the redundant link status
- Step: Ensure the remote hands are dispatched to the site

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to environmental factors	High
Line card crashes due to software issues or hardware malfunctions	Medium
Power supply redundancy loss due to hardware failure or environmental factors	Low

Diagnostic Commands & Expected Outputs

```
Command: show chassis environment, Expected Output: Chassis temperature, humidity, and power supply status, Description: This command provides information on the chassis environment, including temperature, humidity, and power supply status, which can help identify potential causes of the failure.

Command: show chassis fpc 0, Expected Output: FPC 0 status, including presence detection module (PDM) status, Description: This command provides detailed information on FPC 0, including the presence
```

```
detection module (PDM) status, which can help identify potential
causes of the failure.

Command: show log messages | match ALARM_4264, Expected Output: Alarm
messages related to ALARM_4264, Description: This command provides a
list of log messages related to ALARM_4264, which can help identify
the root cause of the failure.
```

Resolution Steps

Step 1: Step: Replace the failed line card, Command: request chassis fpc replace 0, Description: This step involves replacing the failed line card with a spare unit to restore network connectivity and prevent further degradation.

Step 2: Step: Verify the presence detection module (PDM) status, Command: show chassis fpc 0 | match PDM, Description: This step involves verifying the presence detection module (PDM) status to ensure it is functioning correctly and not contributing to the failure.

Step 3: Step: Power cycle the failed node, Command: request system power-cycling, Description: This step involves power cycling the failed node to clear any software issues and reset the network connectivity.

Post-Check Verification

```
Command: show chassis fpc 0, Expected Output: FPC 0 status, including
presence detection module (PDM) status, Description: This command
provides a confirmation of the FPC 0 status, including the presence
detection module (PDM) status, after the resolution steps.

Command: show log messages | match ALARM_4264, Expected Output: Alarm
messages related to ALARM_4264, Description: This command provides
confirmation that the alarm messages related to ALARM_4264 have been
cleared after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: load complete, Description: This step involves restoring the original configuration to revert any changes made during the resolution steps and ensure the network returns to its previous state.

- Step: Revert the failed line card, Command: request chassis fpc revert 0, Description: This step involves reverting the failed line card to its original state and restoring the original configuration to ensure the network returns to its previous state.